

Injection Moulding Guide

A. Mould Geometry

- Design a two-part mould, split along the cup's symmetry plane, typically in a vertical orientation.
- Include a short sprue or funnel at the cup stem to serve as the silicone injection channel.
- Add 2–3 small vent holes (approximately 0.5 mm in diameter) near the rim or other locations where air is likely to become trapped.
- Incorporate alignment keys, such as 3 mm hemispherical features, to ensure accurate positioning of the mould halves.
- For complex geometries, consider additional mould separations to facilitate demoulding. Ensure that all internal mould components can be securely fixed to prevent displacement during casting.

B. Material Selection and 3D Printing

- Select a filament with sufficient dimensional stability and temperature resistance for the intended curing process. PLA, PETG, ABS, or ASA may be suitable for laboratory moulds, depending on the required temperature.
- Infill: 15%
- Layer height: 0.15–0.2 mm.
- Wall lines: 5.

Surface finishing

1. Sand the mould cavity to reduce visible layer lines and surface irregularities, using progressively finer abrasive paper where accessible.
2. If required, apply a thin, compatible epoxy coating to seal the printed surface and improve smoothness.
3. Allow the coating to cure fully according to the manufacturer's instructions before casting.
4. Inspect the cavity for coating defects, blocked vents, or dimensional changes.

Surface finishing can reduce the transfer of printing texture to the silicone and improve the quality of the final prototype.

C. Mould Preparation

1. Clean the mould halves thoroughly with soap.
2. Rinse and dry the mould completely, ensuring that no cleaning residues remain.
3. Assemble the mould and secure all internal components.
4. Clamp the mould halves using bolts, a clamp frame, or another suitable fixture.
5. Check that the parting line is properly closed and that the sprue and vent channels remain unobstructed.

D. Silicone Mixing and Degassing

1. Determine the correct mixing ratio from the silicone manufacturer's technical data sheet. The ratio may vary between materials and must not be assumed to be 1:1.
2. Prepare the required quantities of Parts A and B.
3. Mix the components slowly and thoroughly, avoiding unnecessary air incorporation.
4. Transfer the mixture to a suitable vacuum container and degas until all bubbles collapsed.
5. Complete mixing, degassing, and casting within the specified pot life.

E. Casting and Manual Injection

General-purpose prototyping silicone

1. Transfer the degassed silicone into a suitable syringe (approximately 20–50 mL, depending on the required volume).
2. Minimize air entrapment during transfer by maintaining a continuous flow and avoiding repeated pouring.
3. Insert the syringe tip into the sprue.
4. Inject the silicone slowly and continuously, allowing the cavity to fill from the injection point toward the vents.
5. Observe the vent openings. The appearance of silicone indicates that the corresponding regions have filled and air has been displaced.
6. If necessary, close the vents using a suitable method (tape or wax) that does not introduce contamination or disturb the mould.
7. Maintain the mould in the required orientation, close the sprue tunnel with a paper towel piece and cure the silicone according to the manufacturer's specified conditions.

Medical-grade silicone (ELASTOSIL® LR 5040/45)

Due to its higher viscosity, the medical-grade silicone used in this thesis could not be injected successfully using the available syringe setup. Instead, the silicone was distributed directly onto the mould surfaces before the mould halves were assembled and clamped.

The manufacturer specifies curing conditions for heated industrial moulds. For the laboratory process used in this thesis, the curing temperature was limited to 100 °C to avoid deformation of the 3D-printed moulds (use ASA). A curing time of approximately 2 hours at 100 °C was established experimentally for this specific setup.

These conditions should be regarded as laboratory-specific and should be verified for any changes in mould material, geometry, silicone batch, or processing equipment.

F. Demoulding and Inspection

1. Allow the mould and silicone to cool sufficiently before handling.
2. Remove the clamps and separate the mould components carefully.
3. Demould the silicone part gradually, paying particular attention to thin membranes, strings, and enclosed features.
4. Remove excess silicone from the sprue, vents, and parting lines.

5. Inspect the prototype for trapped air, incomplete filling, tears, surface defects, and dimensional inconsistencies.
6. Record any manufacturing defects and their likely causes to support further mould and process optimization.

Note: This procedure describes laboratory prototype manufacturing. The use of medical-grade silicone alone does not make the resulting prototype a certified medical device. Medical-device production would require additional controls, validated manufacturing processes, biocompatibility assessment, and appropriate quality-management documentation.